



MAINS VOLTAGE on the primary side -- 6.4 mm creepage to everything else
Generated by PCB/acdc_converter/design.py -- edit the script, not the sheet
CERN-OHL-W-2.0

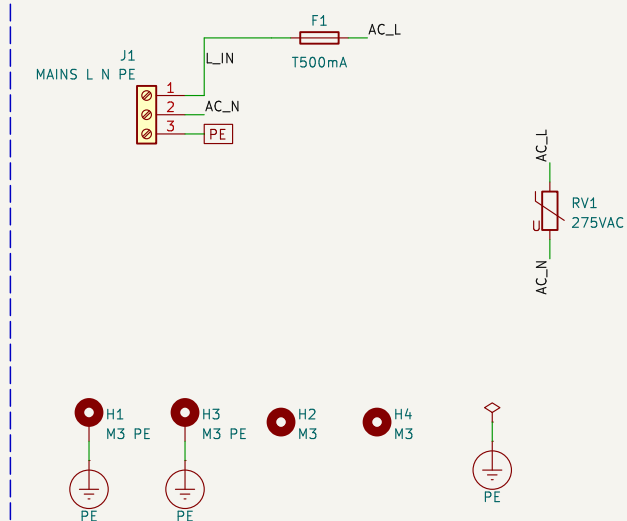
ELARA -- ELF Atmospheric Radio Analyser

Sheet: /
File: acdc_converter.kicad_sch

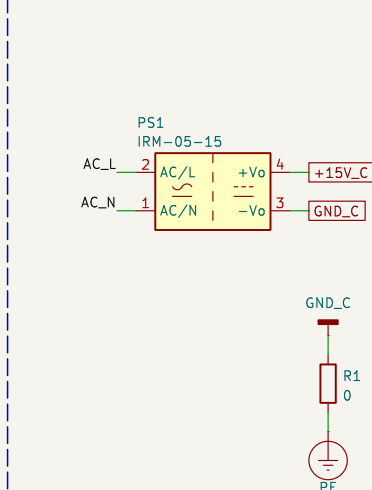
Title: ELARA PSU

Size: A4	Date: 2026-09-30	Rev: 0.2
KiCad E.D.A. 9.0.9	Id: 1/2	

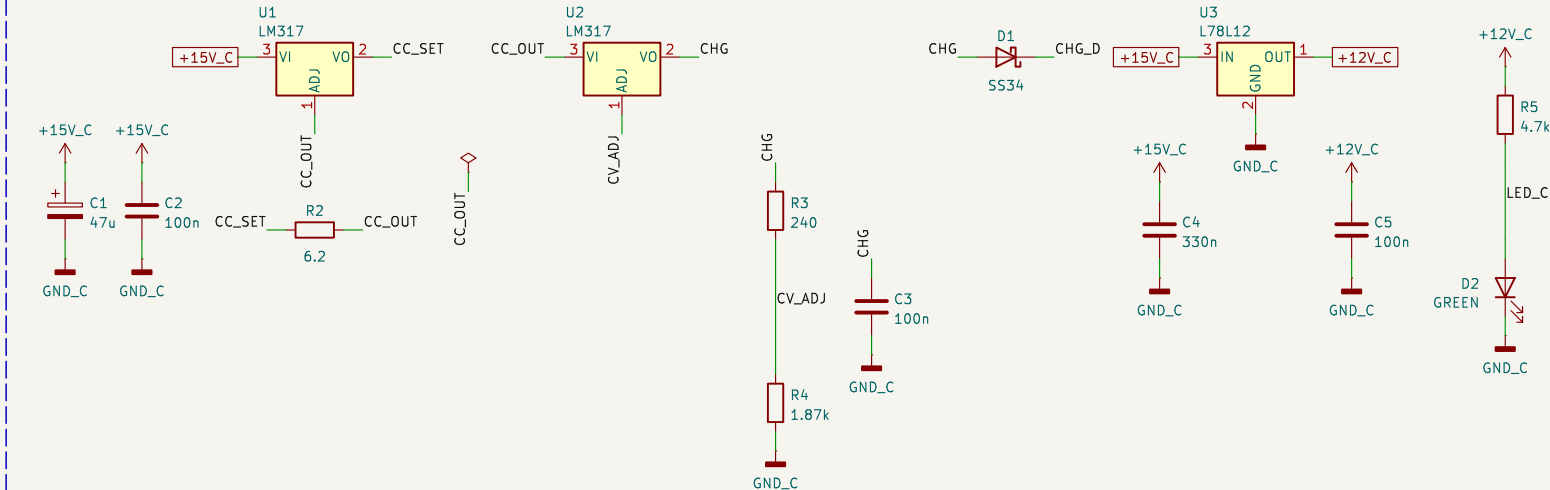
MAINS (primary)



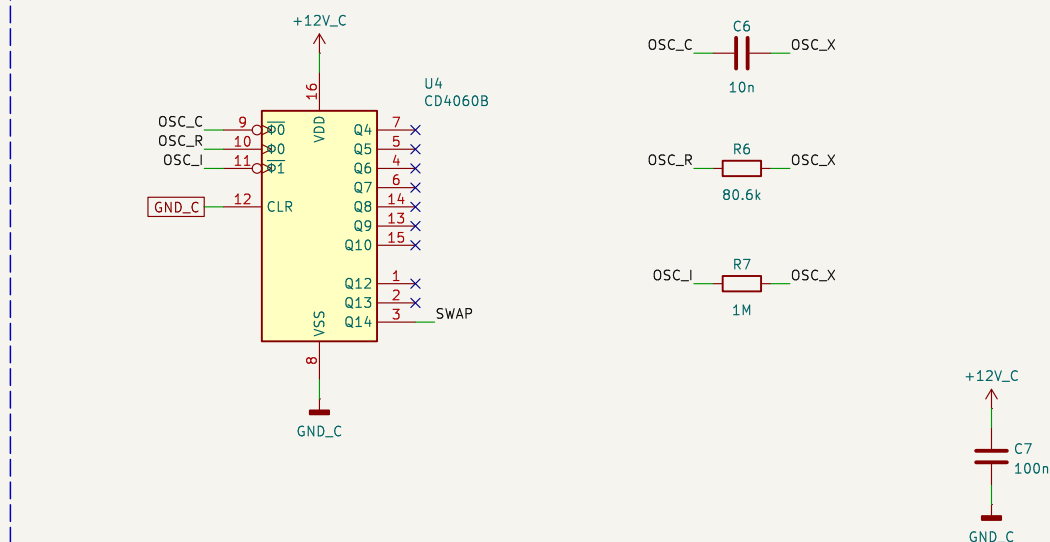
AC-DC 15 V



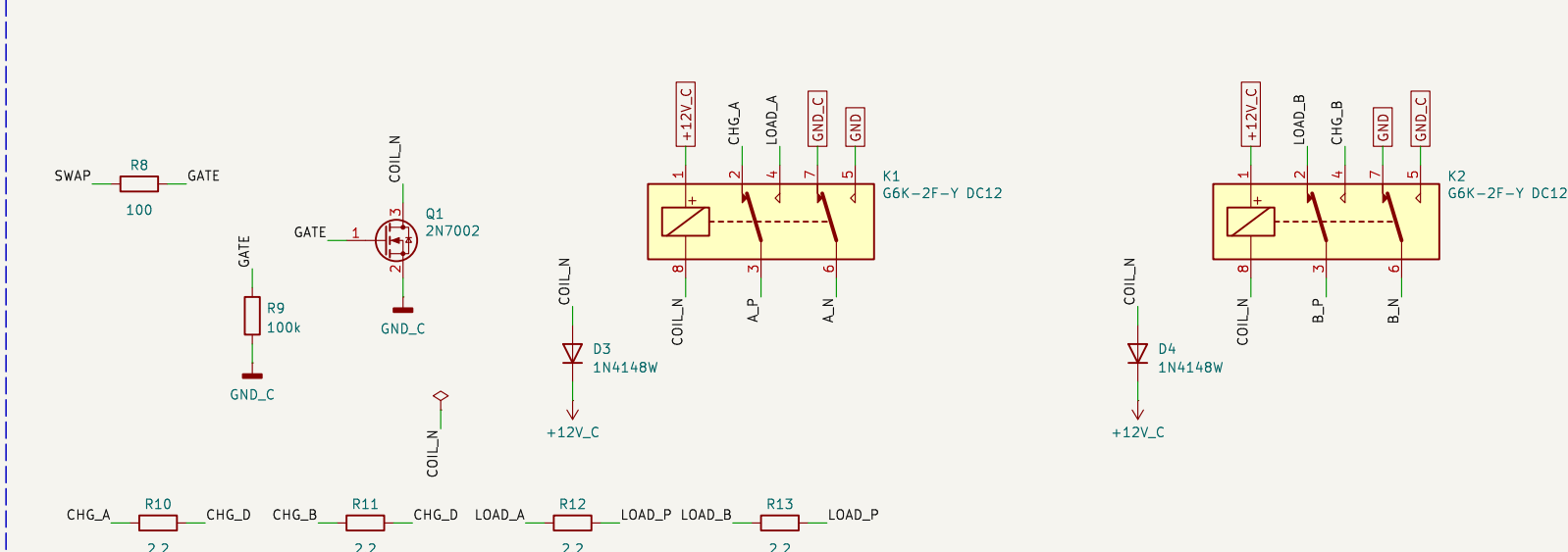
CHARGER (GND_C) CC 0.2 A -> CV 10.9 V



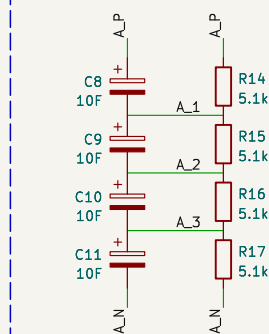
SWAP TIMER Q14 toggles every ~15 s



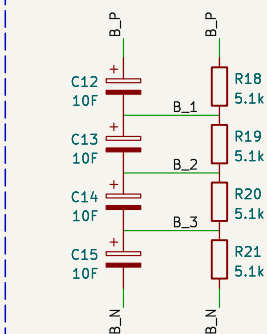
BUCKET SWAP (form C: break before make)



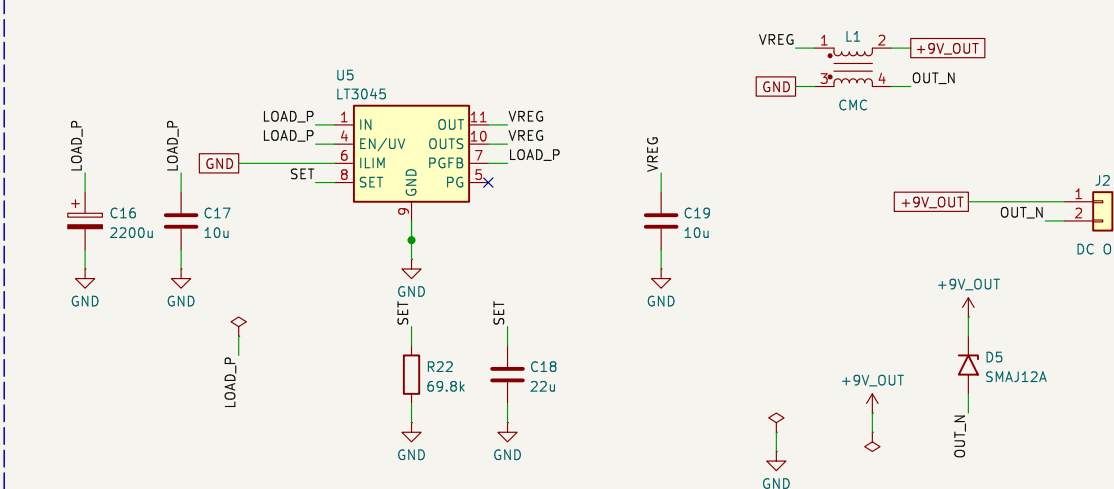
BUCKET A 2.5 F



BUCKET B 2.5 F



RECEIVER SIDE LT3045 6.98 V



Buckets top out at ~10.3 V (two LM317 drops + SS34).
15 s at ~0.11 A droops ~0.7 V; the LT3045 at 6.98 V keeps
>= 0.5 V headroom in the worst corner (simulations/psu).
Swap rate 33 mHz, far below the ELF band.
+9V_OUT carries the 6.98 V output (net name kept from rev 0.1).